



Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)



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WEEKLY LESSON PLAN

Department: Computer Science and Engineering

Year / Semester: 2025-26 / Winter

Course Code / Course Name: 10212CS292 / Applied Programming Skills

Unit V: Algorithm Design Techniques				
Course Outcome: Develop algorithmic solutions using programming strategies for optimization problems.				
Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 9	Session 41	Divide and Conquer pattern	Task 71: Majority Element Task 72: Reverse Bits Task 73: Longest Nice Substring	Leet Code
	Session 42	Merge sort for counting inversions	Task 74: Count of Smaller Numbers After Self	
	Session 43	Quick select (Kth element)	Task 75: K Closest Points to Origin	
	Session 44	Dynamic Programming	Task 76: Climbing Stairs	
	Session 45	One Dimensional (house robber, climbing stairs)	Task 77: House Robber Task 78: Min Cost Climbing Stairs	

Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 10	Session 46	Two Dimensional (grid paths)	Task 79: Unique Paths	Leet Code
	Session 47	Subset Dynamic Programming (0/1 knapsack style)	Task 80: Ones and Zeroes Task 81: Coin Change	
	Session 48	Greedy interval scheduling	Task 82: Minimum Number of Arrows to Burst Balloons	
	Session 49	Backtracking (subsets, permutations, combinations)	Task 83: Subsets	
	Session 50		Task 84: Permutations Task 85: Combinations	